Intraocular Recurrence of Uveal Melanoma after Proton Beam Irradiation

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The authors evaluated tumor recurrence as an end point in 1077 uveal melanoma patients treated by proton beam irradiation between 1975 and 1987, with a mean follow-up of 4.0 years. Twenty tumors (1.9%) exhibited definitive growth between 4 months and 66 months after irradiation (median, 19 months): 10 were marginal recurrences, 5 were ring melanomas, 3 were uncontrolled tumors, and 2 were extrascleral extensions. Ten of these eyes were enucleated without further intervention, 2 patients died of metastases soon after the growth was detected, and 8 patients had further conservative therapy. In 5 other patients, tumor growth was suspected but unconfirmed and the eye was enucleated elsewhere; the 5-year probability of local tumor control based on 25 recurrences was 97% ± 1%. There was a nonsignificant increase in the rate of metastases among patients with documented tumor recurrence (adjusted relative risk, 1.5; 95% confidence interval, 0.66 to 3.4). Results suggest that recurrence of uveal melanomas treated by proton beam therapy is uncommon, but that treatment failure may increase risk of death from metastasis. *Ophthalmology 1992;99:760–766*

Effectiveness of irradiation in the treatment of uveal melanoma can be gauged in terms of rates of complications and retention of the eye, visual results, local tumor control, and patient survival. Studies of patients treated by proton beam irradiation have demonstrated that useful vision is often retained after irradiation,[1,2] with best results observed when the tumor is situated more than 2 disc diameters from the macula and optic disc.[1] Retention of the eye 5 years after proton therapy is achieved in 90% of patients, and factors predicting eye loss after proton beam irradiation have been delineated.[3] The 5-year cumulative probability of developing metastases after proton beam irradiation is 20%,[4,5] with clinical prognostic factors for metastases comparable with those found for patients treated by enucleation.[4,5] Results of nonrandomized studies suggest no appreciable differences in survival after proton beam irradiation compared with enucleation.[6] Preliminary studies involving patients treated at the Harvard Cyclotron suggest that local treatment failure is an uncommon event after proton therapy.[7]

Tumor recurrence is an important end point because it signals the presence of active tumor and therefore the necessity for further management. Continued tumor growth has been reported to occur in 4% to 18% of tumors treated by plaque radiotherapy.[8–14] Results after charged particle irradiation at other centers also have been reported:[15,16] tumor growth occurred in 6 of 284 tumors (2.1%) treated with variable doses of helium ions (50 to 80 Gy),[15] and in 12 of the first 63 tumors (19%) treated with proton beam irradiation in the Soviet Union.[16] With a few exceptions,[7,13] most studies to date have not eval-

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